Math 431 – Spring 2026

Homework 2 – Solutions

1.12 The sample space and probability measure for this experiment were described in the solution to 1.10 in HW1 : $\Omega = \{1, 2 \dots\} \cup \{\infty\}$ with $P(\{k\}) = (\frac{5}{6})^{k-1}\left(\frac{1}{6}\right)$ for positive integers k, and $P(\{\infty\}) = 0$.

(a) $P(\text{need at most 3 rolls}) = P(1) + P(2) + P(3) = \frac{1}{6}\big(1 + \frac{5}{6} + (\frac{5}{6})^2\big) = \frac{91}{216}$.

(b)

$$P(\text{even number of rolls}) = \sum_{m=1}^{\infty} P(\{2m\}) = \sum_{m=1}^{\infty} (\tfrac{5}{6})^{2m-1}\tfrac{1}{6} = \tfrac{1}{5}\sum_{m=1}^{\infty}(\tfrac{25}{36})^m$$
$$= \tfrac{1}{5} \cdot \frac{\frac{25}{36}}{1 - \frac{25}{36}} = \tfrac{5}{11}.$$

1.14 From the inclusion-exclusion principle we get

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.4 + 0.7 - P(AB) = 1.1 - P(AB).$$

Rearranging this we get $P(AB) = 1.1 - P(A \cup B)$.

Since $P(A \cup B)$ is a probability, it is at most 1, so

$$P(AB) = 1.1 - P(A \cup B) \geq 1.1 - 1 = 0.1.$$

On the other hand, $B \subset A \cup B$ so $P(A \cup B) \geq P(B) = 0.7$ which gives

$$P(AB) = 1.1 - P(A \cup B) \leq 1.1 - 0.7 = 0.4.$$

Putting these together we get $0.1 \leq P(AB) \leq 0.4$.

1.16 If we see only heads, I win \$5. If we see 4 heads, I win \$3. If we see 3 heads, I win \$1. If we see 2 heads, I "win" -\$1. If we see 1 heads, I "win" -\$3. Finally, if we see 0 heads, then I "win" -\$5. Thus, the possible values of X are $\{-5, -3, -1, 1, 3, 5\}$. The sample space for the 5 coin flips is $\Omega = \{(x_1, \dots, x_5) : x_i \in \{H, T\}\}$ with $\#\Omega = 2^5$. Each individual outcome $(x_1, \dots, x_5)$ of five flips has probability 2^{-5}.

Let $k \in \{0, 1, \dots, 5\}$. To calculate the probability of exactly k heads we need to count how many five-flip outcomes yield exactly k heads. The answer is $\binom{5}{k}$, the number of ways of specifying which of the five flips are heads. Hence

$$P(\text{precisely } k \text{ heads})$$
$$= \frac{\#\text{ ways to select } k \text{ slots from the 5 for the } k \text{ heads}}{2^5} = \binom{5}{k}2^{-5}.$$

Thus,

$$\begin{aligned}
P(X=-5) &= P(0 \text{ heads}) = 2^{-5} \\
P(X=-3) &= P(1 \text{ heads}) = 5 \cdot 2^{-5} \\
P(X=-1) &= P(2 \text{ heads}) = \binom{5}{2} \cdot 2^{-5} \\
P(X=1) &= P(3 \text{ heads}) = \binom{5}{3} \cdot 2^{-5} \\
P(X=3) &= P(4 \text{ heads}) = \binom{5}{4} \cdot 2^{-5} \\
P(X=5) &= P(5 \text{ heads}) = \binom{5}{5} 2^{-5}.
\end{aligned}$$

1.18 The possible values of X are $\{3,4,5\}$ as these are the possible lengths of the words. The probability mass function is

$$\begin{aligned}
P(X=3) &= P(\text{we chose one of the letters of ARE}) = \frac{3}{16} \\
P(X=4) &= P(\text{we chose one of the letters of SOME or DOGS}) = \frac{8}{16} = \frac{1}{2} \\
P(X=5) &= P(\text{we chose one of the letters of BROWN}) = \frac{5}{16}.
\end{aligned}$$

1.32 Here is one way to determine the number of ways to be dealt a full house. We take as our sample space the set of 5-element subsets of the deck of cards:

$$\Omega = \{\{x_1, \ldots, x_5\} : x_i \in \{\text{deck of } 52\}, x_i \neq x_j \text{ if } i \neq j\}.$$

Note that $\#\Omega = \binom{52}{5}$ (and that we are using a model where we do not keep track of the order of the cards).

Let A be the event that we get a full house (three cards with the same face value and two cards with another common face value). We need $\#A$, i.e. we need to count the number of ways to get a full house. We do this by 'constructing' a full house, and keeping track of the number of choices during the process.

First, choose the face value for the 3 cards that share a face value. There are 13 options. Then select 3 of the 4 suits for this face value. There are $\binom{4}{3}$ ways to do that. We now have the three of a kind selected. Next, choose another face value for the remaining two cards from the remaining 12 face values. Then select 2 of the 4 suits for this face value. There are $\binom{4}{2}$ ways to do that. By the multiplication rule we conclude that there are

$$13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}$$

ways to be dealt a full house. Since there are a total of $\binom{52}{5}$ poker hands, the probability is

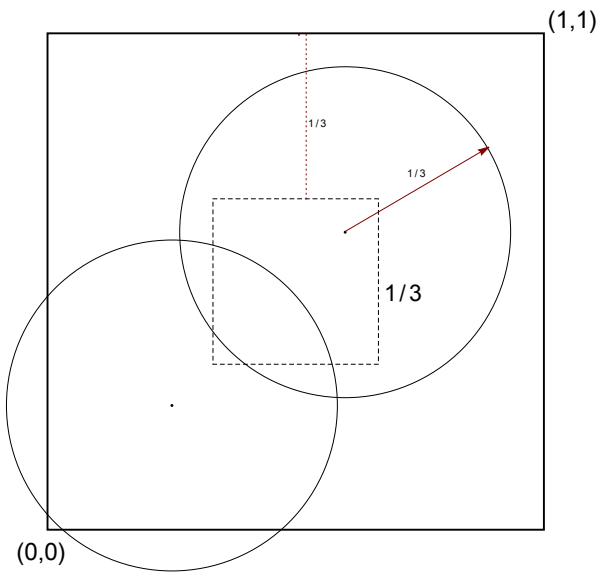
$$P(\text{full house}) = \frac{13 \cdot 12 \cdot \binom{4}{3}\binom{4}{2}}{\binom{52}{5}} \approx 0.00144.$$

1.34 Our sample space Ω is the points of the unit square $[0,1]^2$, we measure the probability using the relative area (which is equal to the area in this case, since the unit square has area 1).

Let A be the event that the circle drawn around the chosen point is completely in the square. The circle of radius of $1/3$ around the random point is completely within the square $[0,1]^2$ if and only if the random point lies within the smaller square with corners $(1/3, 1/3)$, $(2/3, 1/3)$, $(2/3, 2/3)$, $(1/3, 2/3)$. Hence A is exactly this smaller square of side length $1/3$.

The unit square has area one and the smaller square has area $1/9$. Consequently

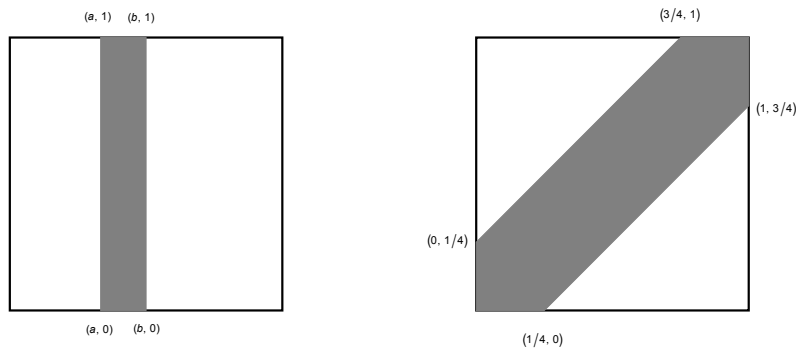
$$P(A) = P(\text{the circle lies inside the unit square}) = \frac{\text{area of } A}{\text{area of } \Omega} = \frac{1/9}{1} = \frac{1}{9}.$$



1.36 Again our sample space Ω is the points of the unit square $[0,1]^2$, we measure the probability using the relative area (which is equal to the area in this case, since the unit square has area 1).

(a) Since (X, Y) is a uniformly random point, probability is proportional to area:

$$\begin{aligned}
&P(a < X < b) \\
&= P(\text{point } (X,Y) \text{ lies in rectangle with vertices } (a,0),(b,0),(b,1),(a,1)) \\
&= \frac{\text{area of rectangle with vertices } (a,0),(b,0),(b,1),(a,1)}{\text{area of square with vertices } (0,0),(1,0),(1,1),(0,1)} \\
&= b - a.
\end{aligned}$$



(b) Let B be the event that $|X - Y| \leq 1/4$, this is the set of points (x, y) that are in the unit square for which the inequality $|x - y| \leq 1/4$ holds. This is the region between the lines $y = x - 1/4$ and $y = x + 1/4$ within the unit square (see the picture above). The area of this region B is $7/16$, this can be seen by by subtracting the areas of the complementary triangles from the unit square. Thus, the desired probability $P(B)$ is also $7/16$ since the unit square has an area of one.

1.40 This is an application of inclusion-exclusion with four events. Below we use some hopefully self-evident summation notation to avoid writing out long sums.

$$\begin{aligned} P(\text{at least one color is repeated exactly twice}) &= P(G \cup R \cup Y \cup W) \\ &= P(G) + P(R) + P(Y) + P(W) - \sum_{\substack{A,B \in \{G,R,Y,W\} \\ A \neq B}} P(AB) \\ &+ \sum_{\substack{A,B,C \in \{G,R,Y,W\} \\ A,B,C \text{ distinct}}} P(ABC) - P(GRYW) \end{aligned}$$

(The first sum contains the probabilities of double intersections, the second the probabilities of triple intersections.)

Next we derive the probabilities that appear in the equation above. The sample space Ω is given by 4-tuples (x_1, x_2, x_3, x_4) where each x_i is from the set $\{\texttt{green}, \texttt{red}, \texttt{yellow}, \texttt{white}\}$. The total number of 4-tuples is $4^4 = 256$.

$$P(G) = P(\text{exactly two greens}) = \frac{\#G}{\#\Omega} = \frac{\binom{4}{2} \cdot 3 \cdot 3}{256} = \frac{27}{128}.$$

The numerator above is derived as follows: there are $\binom{4}{2}$ ways to pick the positions of the two greens in the 4-tuple. For both of the remaining two positions we have 3 colors to choose from. By the same reasoning, $P(G) = P(R) = P(Y) = P(W) = \frac{27}{128}$.

An event of type AB above means that the four draws yielded two balls of color `a` and two balls of color `b`, where `a` and `b` are two distinct particular colors. The number of 4-tuples in the event AB is $\binom{4}{2} = 6$. We can even list them easily. Here they are in lexicogaphic order:

$$\texttt{aabb}, \texttt{abab}, \texttt{abba}, \texttt{baab}, \texttt{baba}, \texttt{bbaa}.$$

Thus for each pair of events A, B we get $P(AB) = 6/256 = 3/128$.

Events of the type ABC are empty because four draws cannot yield three different colors that each appear exactly twice. For the same reason $GRYW = \varnothing$.

Putting everything together gives

$$\begin{aligned} &P(\text{at least one color is repeated exactly twice}) \\ &= 4 \cdot \frac{27}{128} - 6 \cdot \frac{3}{128} = \frac{45}{64} \approx 0.7031. \end{aligned}$$

Another solution: You can also get the answer by counting the favorable outcomes directly, although some care is needed. If we have at least one color repeated exactly twice then we have one or two colors that are repeated twice. (We cannot have more than two because we only have four picks.) Let A_1 be the event that there is exactly one color that is repeated twice (the for picks are of the form $\mathtt{a}, \mathtt{a}, \mathtt{b}, \mathtt{c}$ *in some order*), and A_2 the event that there are exactly two colors that are repeated twice (the for picks are of the form $\mathtt{a}, \mathtt{a}, \mathtt{b}, \mathtt{b}$ *in some order*).

We can uniquely describe an element (x_1, x_2, x_3, x_4) of A_2 by giving the color of x_1 (4 choices), the position of the other ball with the same color as x_1 (3 choices), and the other color that shows up besides x_1's color (3 choices). This gives $\#A_2 = 4 \cdot 3 \cdot 3 = 36$.

An element (x_1, x_2, x_3, x_4) of A_1 can be uniquely described by identifying the repeated color (4 choices), the positions where this color shows up ($\binom{4}{2} = 6$ choices), and choosing the colors of the remaining two balls in order (3 and 2 choices, respectively). This gives $\#A_1 = 4 \cdot \binom{4}{2} \cdot 3 \cdot 2 = 144$. Hence

$$\#\text{at least one color is repeated exactly twice} = 36 + 144 = 180$$

and the corresponding probability is $\frac{180}{4^4} = \frac{45}{64}$.